

13-mavzu. O'zg armas koefitsientli chiziqli differensial tenglamalar. (2-soat).

Reja:

1. O'zg armas koefitsientli chiziqli differensial tenglamalar.
2. Bir jinsli bo'lmagan o'zgaras koefitsientli chiziqli differensial tenglamalar va ularning xususiy yechimlarini topish usullari.
3. Eyler tenglamasi.
4. Bazis yechimlar, fundamental Sistema, Eyler tenglamasi

Bu paragrafda quyidagi n - tartibli chiziqli uzluksiz koefitsientli bir jinsli differensial tenglama yechimining tuzilishini o'rganamiz:

$$L[y] = 0, \quad (1)$$

bunda

$$L[y] = y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y$$

va $\{a_{n-1}(x), \dots, a_1(x), a_0(x)\} \subset C(I)$.

n - tartibli chiziqli bir jinsli differensial tenglama (1) ning n dona chiziqli erkli yechimlari bu **tenglamaning bazis yechimlari** yoki **yechimlarning fundamental sistemasi** deb ataladi.

Teorema. Ixtiyoriy chiziqli bir jinsli (1) differensial tenglamaning bazis yechimlari mavjud va uning har qanday yechimi biror bazis yechimlarining chiziqli kombinatsiyasidan iborat, ya'ni $\dim V_n = n$.

□Quyidagi n guruh boshlang'ich shartlarni qarayluk ($x_0 \in I$):

$$y(x_0) = 1, y'(x_0) = 0, y''(x_0) = 0, \dots, y^{(n-2)}(x_0) = 0, y^{(n-1)}(x_0) = 0; \quad (2)$$

$$y(x_0) = 0, y'(x_0) = 1, y''(x_0) = 0, \dots, y^{(n-2)}(x_0) = 0, y^{(n-1)}(x_0) = 0; \quad (3)$$

$$\dots$$

$$y(x_0) = 0, y'(x_0) = 0, y''(x_0) = 0, \dots, y^{(n-2)}(x_0) = 0, y^{(n-1)}(x_0) = 1; \quad (n)$$

Bu yechimlarni mos ravishda $y = \varphi_1(x), y = \varphi_2(x), \dots, y = \varphi_n(x)$ bilan belgilaylik. Shunday qilib,

$$L[\varphi_i(x)] = 0, \quad \varphi_i^{(i-1)}(x_0) = 1, \varphi_i^{(j)}(x_0) = 0, j \neq i-1, i, j = \overline{1, n} \text{ (ya'ni } \varphi_i^{(j)}(x_0) = \delta_{i,j+1}). \quad (4)$$

Bu $y = \varphi_1(x), y = \varphi_2(x), \dots, y = \varphi_n(x)$ yechimlar chiziqli erkli, chunki ularning vronskiani $x = x_0$ nuqtada (3) ga ko'ra, ravshanki, birga teng. Demak, ular bazis yechimlarni tashkil etadi.

Endi ixtiyoriy $\varphi_1(x), \varphi_2(x), \dots, \varphi_n(x)$ bazis yechimlar berilgan bo'lsin. Ular yuqorida qurilgan bazis yechimlardan farqli bo'lishi ham mumkin. Ixtiyoriy $y = y(x)$ yechim shu bazis yechimlarning chiziqli kombinatsiyasidan iborat ekanligini ko'rsatishimiz kerak. $\varphi_1(x), \varphi_2(x), \dots, \varphi_n(x)$ chiziqli erkli yechimlar bo'lgani uchun ularning vronskiani ixtiyoriy $x_0 \in I$ nuqtada noldan farqli. Demak, ushbu

$$c_1\varphi_1(x_0) + c_2\varphi_2(x_0) + \dots + c_n\varphi_n(x_0) = y(x_0),$$

$$c_1\varphi_1'(x_0) + c_2\varphi_2'(x_0) + \dots + c_n\varphi_n'(x_0) = y'(x_0),$$

$$\dots$$

$$c_1\varphi_1^{(n-1)}(x_0) + c_2\varphi_2^{(n-1)}(x_0) + \dots + c_n\varphi_n^{(n-1)}(x_0) = y^{(n-1)}(x_0)$$

algebraik sistemani qanoatlantiruvchi yagona $c_1, c_2, \dots, c_n$ yechim mavjud. Ana shu $c_1, c_2, \dots, c_n$ larga ko'ra $\varphi(x) = c_1\varphi_1(x) + c_2\varphi_2(x) + \dots + c_n\varphi_n(x)$ funksiyani tuzaylik. Yechimlarning chiziqli kombinatsiyasi sifatida $\varphi(x)$ ham yechim. Bundan tashqari, $c_1, c_2, \dots, c_n$ larning tanlanishiga ko'ra $\varphi(x_0) = y(x_0), \varphi'(x_0) = y'(x_0), \dots, \varphi^{(n-1)}(x_0) = y^{(n-1)}(x_0)$. Demak, $\varphi(x)$ va $y(x)$ yechimlar bir xil boshlang'ich shartlarni qanoatlantiradi. Yechimning yagonalik xossasiga ko'ra $\varphi(x) \equiv y(x)$, ya'ni $y(x) = c_1\varphi_1(x) + c_2\varphi_2(x) + \dots + c_n\varphi_n(x)$. □

Isbotlangan teoremda $L[\circ]$ operatorning koeffitsientlari uzluksiz ekanligi ahamiyatli. Agar $L[\circ]$ operatorning koeffitsientlari uzluksiz bo'lmasa, teoremaning xulosasi noto'g'ri bo'lishi mumkin. Buni quyidagi misol asoslaydi.

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$$y'' - a(x)y' = 0, \quad a(x) = \begin{cases} \frac{2}{x}, & \text{agar } x \neq 0 \text{ bo'lsa,} \\ 0, & \text{agar } x = 0 \text{ bo'lsa,} \end{cases}$$

ikkinchi tartibli (uzluksiz bo'lmagan koeffitsientli) chiziqli differensial tenglamani qaraylik. Bu tenglama $(-\infty; +\infty)$ oraliqda aniqlangan $y_1 = 1, y_2 = x^3, y_3 = |x|^3$ yechimlarga ega (tekshirib ko'ring). Bu yechimlar chiziqli erkli. Haqiqatan ham, faraz qilaylik,

$$\lambda_1 + \lambda_2 x^3 + \lambda_3 |x|^3 = 0, \quad x \in \mathbb{R},$$

bo'lsin. U holda

$$\lambda_1 + \lambda_2 x^3 + \lambda_3 x^3 = 0, \quad x \geq 0,$$

$$\lambda_1 + \lambda_2 x^3 - \lambda_3 x^3 = 0, \quad x \leq 0,$$

bo'ladi. Bu ayniyatlardan $\lambda_1 = 0, \lambda_2 + \lambda_3 = 0, \lambda_2 - \lambda_3 = 0$, ya'ni $\lambda_1 = \lambda_2 = \lambda_3 = 0$

ekanligini topamiz. Demak, qaralayotgan $y_1 = 1, y_2 = x^3, y_3 = |x|^3$ funksiyalarning faqat trivial chiziqli kombinatsiyasigina aynan nolga teng, ya'ni ular chiziqli erkli. Shunday qilib, berilgan ikkinchi tartibli chiziqli differensial tenglama uch dona chiziqli erkli yechimga ega ekan. Agar $a(x)$ koeffitsient $(-\infty; +\infty)$ oraliqda uzluksiz bo'lganda edi, bu differensial tenglama ikkita chiziqli erkli yechimga ega bo'lib, ixtiyoriy uchinchi yechim shu ikki yechimning chiziqli kombinatsiyasidan iborat bo'lar edi. □

Masalalar

1. $y'' + \omega^2 y = 0, \omega = \text{const} > 0$, tenglamaning umumiy yechimi

$y = c_1 \cos \omega x + c_2 \sin \omega x$ formula bilan berilishini ko'rsating.

2. Ushbu Eyler tenglamasini qaraylik. $\lambda^2 + (p-1)\lambda + \beta = 0$ kvadrat tenglamaning diskriminantini $D = (p-1)^2 - 4\beta$, ildizlarini esa λ_1, λ_2 bilan belgilaylik. Eyler tenglamasining umumiy yechimi

$$y = c_1 x^{\lambda_1} + c_2 x^{\lambda_2}, \quad \text{agar } D > 0 \text{ bo'lsa;}$$

$$y = (c_1 + c_2 \ln x) x^{\lambda_1}, \quad \text{agar } D = 0 \text{ bo'lsa;}$$

$$y = (c_1 \cos(\text{Im} \lambda_1 \ln x) + c_2 \sin(\text{Im} \lambda_1 \ln x)) x^{\text{Re} \lambda_1}, \quad \text{agar } D < 0 \text{ bo'lsa,}$$

ko`rinishda tasvirlanadi. Shuni isbotlang.

Biz yuqorida n - tartibli chiziqli bir jinsli differensial tenglama $L[y]=0$ ning bazis yechimlari mavjud ekanligini asoslagan edik. Endi teskari masala bilan shug'ullanamiz, ya'ni bazis yechimlariga ko'ra mos differensial tenglamani tiklash masalasini o'rganamiz.

Mavzuni mustahkamlash uchun savollar

1. O'zg armas koefitsientli chiziqli differensial tenglamalar deb nimaga aytiladi?
2. Bir jinsli bo'lmagan o'zgaras koefitsientli chiziqli differensial tenglama deb nimaga aytiladi?
3. Eyler tenglamasi deb nimaga aytiladi?